Scientific Notation

Name _____ **Date** _____ Calculator allowed **Version 1**

KEY WORDS scientific notation · standard form · power of 10 · exponent · ordinary number

A Powers of 10 ●○○

ANSWER ON THE SHEET

1. Write as an ordinary number.

a) 10^3 = _____ **b)** 10^6 = _____ **c)** 10^{-1} = _____ **d)** 10^{-3} = _____

2. Write as a power of 10.

a) 100 = _____ **b)** 10,000 = _____ **c)** 0.01 = _____ **d)** 0.0001 = _____

B Writing large numbers ●○○

ANSWER ON THE SHEET

> **Remember:** Scientific notation is $a \times 10^n$, where a is at least 1 and less than 10.

3. Write in scientific notation.

a) 4,000 = _____ **b)** 62,000 = _____ **c)** 750,000 = _____

d) 3,400,000 = _____ **e)** 81,500 = _____ **f)** 9,060,000 = _____

g) 120 = _____ **h)** 55.5 = _____ **i)** 23 million = _____

C Writing small numbers ●●○

ANSWER ON THE SHEET

> ⚠ **Watch out:** A number smaller than 1 has a negative power of 10. Count how many places the decimal point moves.

4. Write in scientific notation.

a) 0.005 = _____ **b)** 0.036 = _____ **c)** 0.00081 = _____

d) 0.4 = _____ **e)** 0.0000275 = _____ **f)** 0.00609 = _____

g) 0.0006 = _____ **h)** 0.0000014 = _____ **i)** 7 thousandths = _____

D Back to ordinary numbers ●●○

ANSWER ON THE SHEET

5. Write as an ordinary number.

a) 3×10^4 = _____ **b)** 5.2×10^3 = _____ **c)** 7.08×10^6 = _____

d) 6×10^{-2} = _____ **e)** 4.5×10^{-4} = _____ **f)** 1.29×10^{-3} = _____

g) 2.5×10^5 = _____ **h)** 9.9×10^1 = _____ **i)** 8.03×10^{-5} = _____

E Putting it right ●●○

ANSWER ON THE SHEET

6. These numbers are not in scientific notation. Write each one correctly.

a) 34×10^3 = _____ **b)** 0.6×10^5 = _____ **c)** 125×10^{-3} = _____ **d)** 0.08×10^{-2} = _____

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F Calculating in scientific notation ●●● WORK IN YOUR BOOK

Remember: Multiply or divide the numbers in front, then use the exponent rules on the powers of 10.

7. Work out. Give each answer in scientific notation.

a) $(2 \times 10^3) \times (4 \times 10^5)$

b) $(3 \times 10^4) \times (3 \times 10^{-2})$

c) $(5 \times 10^6) \times (4 \times 10^3)$

d) $(6 \times 10^5) \times (1.5 \times 10^3)$

e) $(8 \times 10^7) \div (2 \times 10^3)$

f) $(9 \times 10^5) \div (3 \times 10^{-2})$

g) $(2 \times 10^4) \div (8 \times 10^6)$

h) $(4.8 \times 10^9) \div (1.2 \times 10^4)$

G Problems ●●● WORK IN YOUR BOOK

8. The distance from Earth to the Sun is about 150,000,000 km. Write this in scientific notation.

9. A human hair is about 0.00007 m wide. Write this in scientific notation.

10. Light travels about 3×10^5 km each second. How far does it travel in 60 seconds? Give your answer in scientific notation.

11. New Zealand has about 5.3×10^6 people and about 2.4×10^7 sheep. About how many sheep is that for each person? Round to 1 decimal place.

12. A sheet of paper is about 1×10^{-4} m thick. How thick is a pile of 500 sheets? Give your answer in scientific notation.

13. The Earth is about 4.5×10^9 years old. Write this as an ordinary number.

★ Extension FOR EARLY FINISHERS

14. Work out $(4 \times 10^5) + (3 \times 10^4)$. Give your answer in scientific notation.

15. Put these in order, smallest first: 3.2×10^4, 8×10^3, 4.1×10^{-2}, 9.9×10^{-3}.

16. Work out $(3 \times 10^4)^2$. Give your answer in scientific notation.

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Scientific Notation

Name _____ **Date** _____ Calculator allowed **Version 2**

KEY WORDS scientific notation · standard form · power of 10 · exponent · ordinary number

A Powers of 10 ●○○ ANSWER ON THE SHEET

1. Write as an ordinary number.

a) 10^4 = _____ **b)** 10^5 = _____ **c)** 10^{-2} = _____ **d)** 10^{-4} = _____

2. Write as a power of 10.

a) 1,000 = _____ **b)** 100,000 = _____ **c)** 0.1 = _____ **d)** 0.001 = _____

B Writing large numbers ●○○ ANSWER ON THE SHEET

Remember: Scientific notation is $a \times 10^n$, where a is at least 1 and less than 10.

3. Write in scientific notation.

a) 7,000 = _____ **b)** 38,000 = _____ **c)** 290,000 = _____

d) 5,100,000 = _____ **e)** 42,500 = _____ **f)** 6,030,000 = _____

g) 460 = _____ **h)** 72.5 = _____ **i)** 45 million = _____

C Writing small numbers ●●○ ANSWER ON THE SHEET

Watch out: A number smaller than 1 has a negative power of 10. Count how many places the decimal point moves.

4. Write in scientific notation.

a) 0.007 = _____ **b)** 0.042 = _____ **c)** 0.00053 = _____

d) 0.9 = _____ **e)** 0.0000185 = _____ **f)** 0.00307 = _____

g) 0.0003 = _____ **h)** 0.0000026 = _____ **i)** 4 hundredths = _____

D Back to ordinary numbers ●●○ ANSWER ON THE SHEET

5. Write as an ordinary number.

a) 8×10^4 = _____ **b)** 2.7×10^3 = _____ **c)** 3.05×10^6 = _____

d) 9×10^{-2} = _____ **e)** 6.1×10^{-4} = _____ **f)** 4.83×10^{-3} = _____

g) 4.5×10^5 = _____ **h)** 6.6×10^1 = _____ **i)** 2.07×10^{-5} = _____

E Putting it right ●●○ ANSWER ON THE SHEET

6. These numbers are not in scientific notation. Write each one correctly.

a) 52×10^4 = _____ **b)** 0.3×10^6 = _____ **c)** 270×10^{-4} = _____ **d)** 0.09×10^{-3} = _____

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F Calculating in scientific notation ●●● WORK IN YOUR BOOK

Remember: Multiply or divide the numbers in front, then use the exponent rules on the powers of 10.

7. Work out. Give each answer in scientific notation.

a) $(3 \times 10^2) \times (2 \times 10^6)$

b) $(4 \times 10^5) \times (2 \times 10^{-3})$

c) $(6 \times 10^4) \times (5 \times 10^5)$

d) $(2.5 \times 10^4) \times (2 \times 10^6)$

e) $(9 \times 10^8) \div (3 \times 10^2)$

f) $(8 \times 10^4) \div (4 \times 10^{-3})$

g) $(3 \times 10^3) \div (6 \times 10^7)$

h) $(7.5 \times 10^8) \div (2.5 \times 10^3)$

G Problems ●●● WORK IN YOUR BOOK

8. The distance from Earth to the Moon is about 384,000 km. Write this in scientific notation.

9. A red blood cell is about 0.000008 m across. Write this in scientific notation.

10. Sound travels about 3.4×10^2 m each second. How far does it travel in 50 seconds? Give your answer in scientific notation.

11. Australia has about 2.7×10^7 people and New Zealand has about 5.3×10^6. About how many times as many people live in Australia? Round to 1 decimal place.

12. A grain of sand has a mass of about 4×10^{-6} kg. Find the mass of 2,000 grains. Give your answer in scientific notation.

13. The universe is about 1.38×10^{10} years old. Write this as an ordinary number.

★ Extension FOR EARLY FINISHERS

14. Work out $(6 \times 10^4) + (5 \times 10^3)$. Give your answer in scientific notation.

15. Put these in order, smallest first: 2.5×10^5, 7×10^4, 6.3×10^{-3}, 1.1×10^{-2}.

16. Work out $(2 \times 10^5)^3$. Give your answer in scientific notation.

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